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Interviewer

2 Could you please briefly introduce yourself and provide a bit of background on your experience working with genomic data?

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Participant

4 Yeah, I'm a fifth year PhD student in the Biomedical and Health Informatics Program at UW, and I first started working with genomic data probably the first year that I was in the program. So I was basically just handed a project that someone else had been working on, and I had the code handed to me as well. I basically just kind of had to jump in at that point and try to figure out what I was doing, which I think was a good learning experience, but there were definitely some frustrating things about it, which we can talk about in a little bit. I wouldn't say I have a ton of experience working with genomic data. Like, most of my research now has been more qualitative. But yeah, I mean, I've done a GWAS. I also worked with just different mammalian genomes during my undergraduate research, so that wasn't
9 human genetics necessarily, but I kind of got familiar with the basics of genetics and working with genetic data before I came to grad school.

4:2 I was basically just ...
First experiences

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Interviewer

6 So you said that you were handed a project when you first joined UW. How did you get started and what educational resources did you use, if any, to kind of get your feet wet with that project?

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Participant

Yeah, so one big thing was all of the code was written in Perl, which is kind of outdated, and my background is in Python or a little bit of Java, but yeah, so I had to learn how to read Perl and kind of understand what was going on in the scripts that I was handed. I also just kind of had to learn how to use our computing cluster. I think Genome Sciences has some good resources that kind of point you in the right direction in terms of little things like using modules and actually just like submitting a job to the cluster and checking the status of your job and figuring out where to put the error logs, you know, what went wrong. Yeah, so the two biggest learning curves were learning the language of the code that was handed to me and then just being able to use the resources for high power computing that we had.

4:1 so one big thi...
First experiences
4:3 I think Genom...
Resources used w...
4:4 so the tw...
First experiences

Interviewer

10 What kind of analysis software did you utilize when working with that project or your most recent genomics project?

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Participant

12 Oh my gosh, it was like three years that I did this. I think I remember the name of it? Oh my gosh, Plink is the big one. And then, I guess in terms of visualization and stuff, I used R and maybe some GWAS packages in R that take the Plink output and then plot those in like Manhattan plots, for example, and QQ plots. I think Plink was the big one that I've used.

4:5 Plink is th...

Specific software t...

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Interviewer

14 From what you remember, what was your workflow like when you were using Plink and then subsequently R?

17 **Participant**

Yeah, it was definitely command line. Again, this was just kind of taking
18 what was given to me, but it was all command line. But I had scripts that automated the command line script. Someone who had worked on the project before me had put together a lot of shell scripts that would take a basic Perl script that he'd written and then create a shell script that just automatically ran many different iterations of the Perl script. So I guess there were different environmental variables that were passed through every iteration of the shell script. Yeah. So I guess that's like a form of automation that I used.

4:6 it was definitely co...

Specific software t...

Participant

Yeah, I tried to automate things as much as I could in terms of if I wanted to tweak something in the GWAS pipeline, it would be relatively easy to like do that. But a lot of times that didn't work out just because I feel like there's a constant cost benefit. Do I want to spend time making this all automated or do I want to just get this done? And it's like, I know I can get
19 it done using brute force methods, but automation would be great in the future. And so I kind of did like a half and half approach where I'm like, okay, there are certain things I truly need to automate or else I will never get it done, and there are other things that I'm just going to kind of hand hold the scripts.

Interviewer

20 Got you. Got you. How did you feel when you were working with Plink and did your feelings change as you worked more and more with that software?

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Participant

Yeah, I mean, I think at first I was a little over my head because I was also just learning about the statistical methods involved in genomic analysis and I didn't really have much of a statistics background coming into the program. And of course, genomic analysis is all statistics, basically. And so I was digging through the Plink documentation and trying to figure out what are all these parameters, what are they doing? It was hard to figure out where I needed to change the Plink commands for my own purposes because I just didn't really understand how it was working yet. As I worked with it more, I don't know if I ever fully understood what was going on. But yeah, I think I became less frustrated as I started. Just kind of becoming familiar with the syntax and stuff.

4:7 I think at first I was a little over ... Cognitive reaction

4:8 I ... Emotional reaction

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Interviewer

24 So other than the documentation that was provided probably on Plink's website, did you have any other questions or concerns about how to utilize the software. Did it slow down your workflow pipeline in any way?

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Participant

26 I think since a lot of the scripts were pre-written, I don't think it slowed it down that much. Thing that took the longest time was just kind of searching through the code, basically finding the needle in the haystack, figuring out, okay, where do I want to change something?

4:9 Thing... Barriers of specific...

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Interviewer

28 Is there anything else that you would just use to describe your overall experience working with Plink?

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Participant

Yeah, let me try to think about that. I think Plink actually is relatively easy to use. The thing that takes the most time is like setting up all your data to get it in a format that Plink will be okay with and then outputting your results and doing all the filtering and stuff. The tough part is just wrangling all of the millions of files that come out of running something like a GWAS. Yeah. I don't know if this gets to like another question you will ask. But one

4:10 ... Benefits of specifi...

4:11 The thin... Barriers of specific...

30 thing that I really found helpful in doing all this work was just keeping an Excel file where I keep track of every single command that I use and all the files that went into that command. And that in its own way is like a sort of automation for me. At least I have it all kind of organized in the list and I know I don't have to think about what I'm going to do next.

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Interviewer

32 Yeah, that's a great idea. It sounds like a command dictionary that you've created for yourself.

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Participant

34 Yeah, exactly.

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Interviewer

36 Let's just spend a little bit more time on some of the positives and negatives of utilizing some of the software you've worked with. So I know
39 we talked about Plink being relatively easy to actually run. Are there any other positives to utilizing Plink and R to do the GWAS analysis?

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40 **Participant**

Yeah, I mean, Plink is really, really fast. I haven't used anything else, but I know that it has been kind of the standard for fast genomic analysis for
41 the past few years. So that was nice. And I mean, it was also very nice to have access to the computing cluster because that absolutely would not have been possible to do the GWAS without it. So I guess that's one thing that made Plink less frustrating than it could have been. Also, we have the RStudio server, and that was kind of a game changer in terms of actually getting the GWAS study done because I'm very visual person. So it was kind of hard to have to interact with all these scripts and files and stuff just by through terminal. It was really helpful to be able to log into our RStudio server and interactively look at the visualizations. So it was totally a game changer to have RStudio in terms of actually working with output data.

4:12 Plink is really, real...
Benefits of specifi...
4:13 Also, we have the RSt...
Benefits of specifi...

Interviewer

Okay, so those are some good positives. I think we touched a little bit on the negatives. Are there any other negatives about utilizing, like, Plink and R in combination?

Participant

42 Yeah, I guess one thing about R, so Plink is really fast and R is really slow. So if you're trying to do any kind of heavy lifting in R, that's going to take
46 a while. But I purposefully didn't try to do any really heavy lifting with R. It was mostly just for data visualization and things like that.

4:14 R is ...

Barriers of specific...

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Interviewer

44 Did utilizing the software that you use influence the cost or upkeep of your experiments in any way? For example, did they create intermediate files that you had to store and pay for?

Participant

I don't know. Actually, that's a good question. I mean, yeah, I'm sure, yeah, I mean, at least the way that you set up the scripts. Plink outputs, like, basically one file per region that you're looking at. So there were hundreds of thousands of files that were output. So yeah that experience alone tells
47 me, okay, even if these are small files, maybe there's storage costs associated with having discrete files for each of these outputs. But also, I think that does pale in comparison to how large, how costly it is to just
48 store the genomic data itself and I tried not to generate any kind of output files that were like, copies, for example, of the bam files or something.

4:15 I'm sure, yeah, I mean, at least ...

Impact on cost/up...

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Interviewer

Did utilizing the software influence any other aspect of your experience being a researcher?

Participant

Oh, yeah, no, I actually took, this isn't software related, but I did take Advanced Human Genetics, like my second quarter or something. I don't think I'd ever even taken a genetics course before that. And that was yeah, I feel like that's been kind of necessary for just understanding the bigger picture of what we're doing. Working with genomic data. I think when you're writing the software and stuff, it's really easy to kind of get lost in the weeds and forget, like, what you're looking at and why you're looking for things. So that's yeah, but then I'm also lucky that I did come into this experience having at least a working knowledge of very common programming languages, languages like Python and R. And, you know, Perl is similar enough to Python that I could learn that I didn't have to take any other programming classes, for example, in order to work with that data.

4:16 I did take Advanced Huma...

Overall impact

Interviewer

52 Got you. And then lastly, other than just some maybe like, code dictionary or command dictionary, do you have any other recommendations for changes or additions that would have aided in your research?

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Participant

54 Yeah, just like documenting all commands has been extremely helpful. Pretty much every project I've done, having some kind of visualization is really important, so maybe that's not important for everyone. Like, some people are really, really comfortable just doing everything through like vi or something on command Line. But I find that really cumbersome. For example, trying to edit text files or something on the command line is so annoying compared to just opening that up in a text editing software or something that we're used to. And you can actually click around, you don't have to press the down arrow for five minutes to get to the area that you want to edit and yeah, so having that real time interactive interface has been really important for me to think of other things that I've used.

4:17 documenting all commands has been e...

Suggested improv...

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Interviewer

56 That makes sense. That makes sense. And lastly, an exploratory question. As the NHGRI and other genomics research organizations move more towards cloud computing, what challenges do you anticipate when conducting research in the cloud?

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Participant

Yeah, I would say the visualization part is huge. I think one thing that cloud computing really relies on are workflows and like the whole workflow description language and that's just a whole other way of kind of setting up your code that most people are not that's just not something that people do on a regular basis. It's much more piecemeal, I would say. Like you kind of like test little bits of it and maybe this is just my personal experience. But I love to like as I'm writing my code. I constantly break it and break it and break it and you can't really do that if you are required to set up this whole script. Wrap it all up in a WDL script or something and then run it and then find your mistake. In Bioformatics, you just kind of learn how to code on the fly and out of necessity sometimes. And a lot of times that's not wrapped up in best practices. Another thing that I've run into doing cloud computing is that it's really, really hard to interact with the terminal sometimes. Like the terminal commands are different. It also depends on how the cloud you're working on. But yeah, as important as it is to be able to interact with your data and code not in the terminal, it's

4:18 I would say the visualization part is huge. I think one thing tha...

Challenges/conce...

58 also really important to just be able to look at your files in the terminal, do little things with them, and that's much harder to do in the cloud.

4:18

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Participant

60 So I don't know if that's something that they're trying to work on, but it looks very different than a Linux environment, for example.

61

Speaker 3

62 Got you.

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Interviewer

Okay, cool. I think those are all my questions. I'm gonna stop the recording now.